

Q1



AARHUS UNIVERSITY

Q2



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ETH zürich

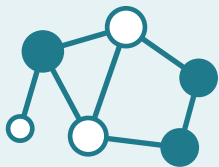
Q3-Q4

ETH zürich

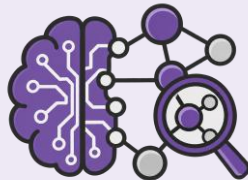
ASSET

Deteriorating bridgeswater ingress,
cracking · corrosion

WP1

Semantic Digital TwinUAS · sensors · reports
→ Knowledge Graph

WP2

Knowledge-grounded AILLM / VLM reasoning
evidence vs. hypotheses

WP3

Immersive Human-AIXR + conversational agent
situational awareness

OUTCOME

Engineering decisioninspect · repair · replace
least intervention**Case studies (Global South):**

Ponte do Funil, Bahia (BR)

Nairobi Expressway (KE)

engineer feedback refines the twin

Q1



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Q2 (joint)



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Q3–Q4

ETH zürich

ASSET

Deteriorating bridges



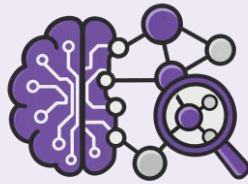
WP1

Semantic Digital Twin



WP2

Knowledge-grounded AI



WP3

Immersive Human–AI



OUTCOME

Engineering decision



Case studies (Global South):

Ponte do Funil, Bahia (BR)

Nairobi Expressway (KE)

*engineer feedback refines the twin*